Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

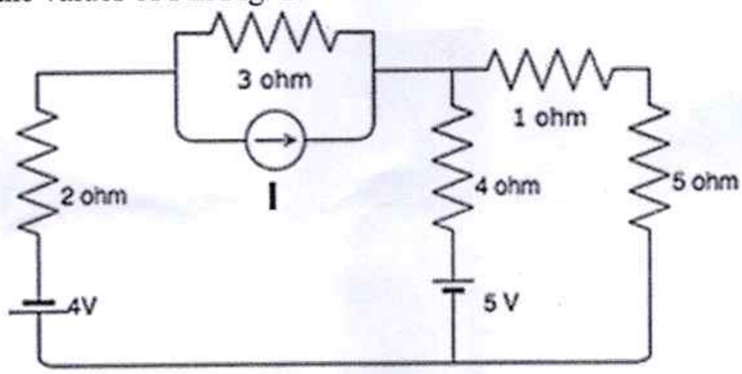
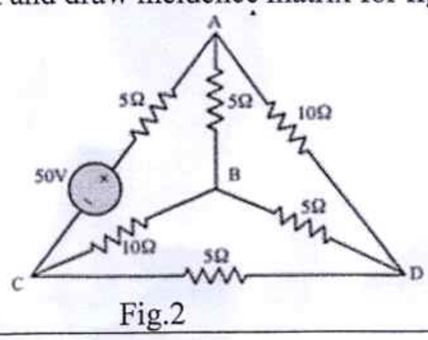
Name of the Program: **B. Tech** Semester: III

Name of the Course: Networks Analysis and Synthesis Course Code: TEC 303

Time: 180 Minutes **Maximum Marks: 100**

Note:

i. All the questions are compulsory.
ii. Answer any two sub questions from a, b and c in each main question.
iii. Total marks for each question are 20 (twenty).
iv. Each sub-question carries 10 marks.

Q1	20 Marks	CO1, CO2
(a)	Determine the values of I in Fig. 1.  Fig. 1	
(b)	How to perform Nodal and Mesh Analysis, Explain it with an Example?	
(c)	Define the following terms,Link (II) Graph (III) Tree (IV) Node (V) Branch (VI) Twig of the following network and draw incidence matrix for fig.2  Fig.2	
Q2	20 Marks	CO2, CO3
(a)	Obtained fundamental incidence, tie-set and cut-set matices for following graph?	

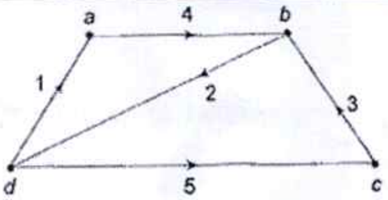


Fig.3

(b) Explain maximum power transfer theorem and Tellegen's theorem with an example?

(c) In the circuit shown in the fig. 4, switch S is in position 1 for a long time and brought to position 2 at time t = 0. Determine the circuit current.

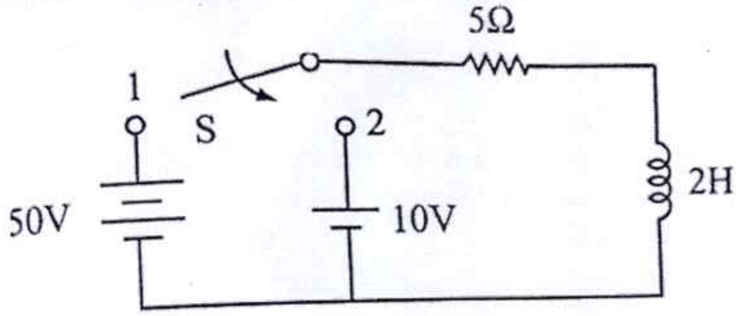


Fig.4

Q3		20 Marks	CO3 CO4

(a) Solve for $i(t)$ for the circuit, given that $V(t) = 10 \sin 5t$ V, $R = 4$ W and $L = 2$ H.

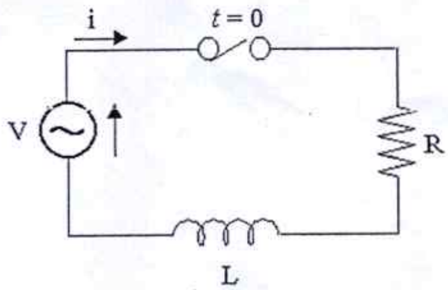


Fig.5

(b) In the circuit given below, the value of the hybrid parameter h_{21} is _________

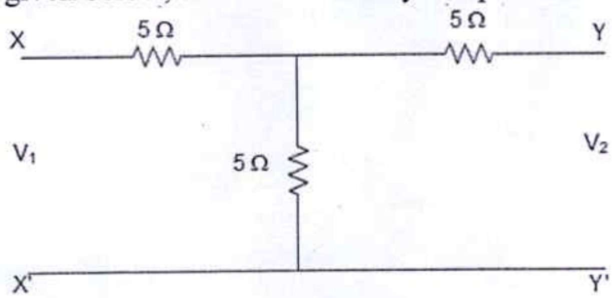


Fig.6

(c) If the following parameters are measured experimentally for a two port network.
i. with output port short-circuited: I1 = 5mA I2= -0.3mAV1 = 20V
ii. with input port short-circuited: I1= -5mA, I2 = 10mA, V2 = 30V then determine the value of Y11 and Y22.

Q4		20 Marks	CO4 CO5

(a) What is equivalent inductance of the following figure (L_{eq})?

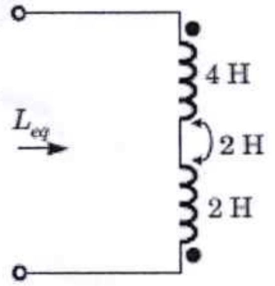
<table>
<tr><td></td><td>

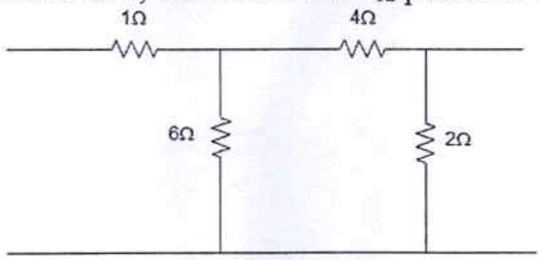
Fig. 7</td><td></td></tr>
<tr><td>(b)</td><td>For the circuit given below, the value of the z_{12} parameter is ___________


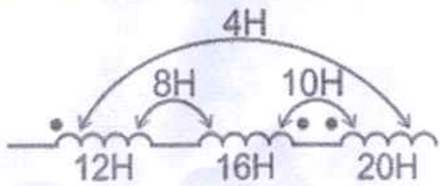
Fig.8</td><td></td></tr>
<tr><td>(c)</td><td>For the three coupled coils in the figure shown below, calculate the total inductance.


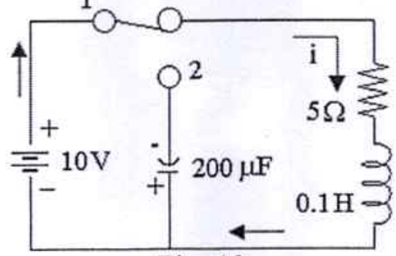
Fig.9</td><td></td></tr>
<tr><td>Q5</td><td>20 Marks</td><td>CO5</td></tr>
<tr><td>(a)</td><td>In the circuit shown, the capacitor has an initial charge of 1 mC and the switch is in position 1 long enough to establish the steady state. The switch is moved from position 1 to 2 at $t = 0$. Obtain the transient current $i(t)$ for $t > 0$.


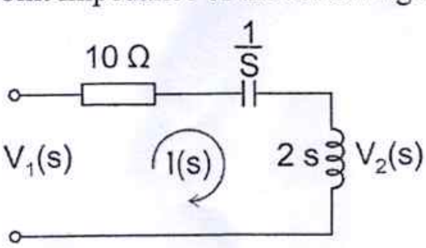
Fig. 10</td><td>CO6</td></tr>
<tr><td>(b)</td><td>The driving point impedance of the following figure will be


Fig. 11</td><td></td></tr>
<tr><td>(c)</td><td>Explain the reciprocity and symmetric condition of all two port networks with an example?</td><td></td></tr>
</table>